16-bit. The higher the sample resolution, the greater the variations in amplitude that can be stored for each sample. This means that aspects such as the loudness of the sound can be recorded more accurately. This will also increase the amount of data that needs to be stored for each sample. This increases the size of the file that stores the sound.

1.4 Measuring data storage

Earlier in the chapter, you learnt that all data need to be converted to digital data to be processed by a computer, and you now know how that is done for different types of data. But how is data measured when it is stored?

COMPUTER SCIENCE IN CONTEXT

There are times when you will need to know the size of a data file, such as when you are attaching a file to an email. Most email providers have a file size limit, so you need to know the size of that file, to know if it can be attached to your email.

Data is measured using different units, which are shown in Table 1.2.

Unit	Description
Bit	This is the smallest measurement for data
Nibble	There are 4 bits in a nibble
Byte	There are 8 bits in a byte
Kibibyte (KiB)	There are 1024 bytes in a Kibibyte
Mebibyte (MiB)	There are 1024 Kibibytes in a Mebibyte
Gibibyte (GiB)	There are 1024 Mebibytes in a Gibibyte
Tebibyte (TiB)	There are 1024 Gibibyte in a Tebibyte
Pebibyte (PiB)	There are 1024 Tebibytes in a Pebibyte
Exbibyte (EiB)	There are 1024 Pebibytes in an Exbibyte

Table 1.2: Data measuring units

Although most computers will calculate the size of a file for you, you need to be able to show that you can do this manually for the exam. This is just for an image file and a sound file.

SKILLS FOCUS 1.3

CALCULATING THE SIZE OF A FILE

There are several pieces of information that you need to know to calculate the size of an image file, these are:

- the number of images in the file
- the resolution of the image
- the colour depth of the image.

CONTINUED

When you know this information, you can perform a simple calculation to get the size of the image file, which is:

width of image × height of image × colour depth of image × number of images in file

Therefore, if you have the following information about an image file:

- 10 images in the file
- image resolution is 100 × 150
- colour depth is 8-bit colour,

you can calculate the size of the image file as follows:

- 100 × 150 = 15,000 (this means there are 15 000 pixels in the image)
- 15 000 × 8 = 120,000 (this means that there are 120 000 bits in each image)
- 120 000 × 10 = 1 200 000 (this means there are 1 200 000 bits in the image file).

You will be asked to calculate the size of an image file to a certain unit of measurement. The answer given here (1 200 000) is the number of bits that are in the image file. If you wanted to know how many bytes are in the image file, you need to divide the final answer by 8, as there are 8 bits in each byte, for example:

1 200 000 ÷ 8 = 150 000 bytes

If you wanted to know how many kibibytes the file was, you need to divide the answer in bytes by 1024, for example:

150 000 ÷ 1024 = 146.5 KiB (to 1 decimal place)

Questions

1 How many times would you need to divide the answer in bytes by 1024 to find the answer in Gibibytes?

2 How many pixels would there be in an image that has a resolution of 1000 × 1500?

There are several pieces of information that you need to know to calculate the size of a sound file, these are:

- the length of the soundtrack
- the sample rate of the soundtrack
- the sample resolution of the soundtrack.

When you know this information, you can perform a simple calculation to get the size of the sound file, which is:

sample rate × sample resolution × length of soundtrack

Therefore, if you have the following information about a sound file:

- length of 30 seconds
- sample rate of 44,100 hz
- sample resolution of 8 bits,

you can calculate the size of the sound file as follows:

- 44 100 × 8 = 352 800 (this means there are 352 800 bits per second)
- 352 800 × 30 = 10 584 000 (this means there are 10 584 000 bits in the sound file).

CONTINUED

You will be asked to calculate the size of a sound file to a certain unit of measurement. The answer given (10 584 000) is the number of bits that are in the sound file. If you wanted to know how many bytes are in the sound file, you need to divide the final answer again by 8, as there are 8 bits in each byte, for example:

$$10\ 584\ 000 \div 8 = 1\ 323\ 000 \text{ bytes}$$

If you wanted to know how many mebibytes the file was, you need to divide the answer in bytes by 1024, then by 1024 again, for example:

$$1\ 323\ 000 \div 1024 \div 1024 = 1.3 \text{ MiB (to 1 decimal place)}$$

1.5 Data compression

You may already have experience of having large files and struggling to have enough storage space to store them. One thing that can be done to reduce the size of a file is **compression**.

Compressing a file will reduce the size of the file that can have several benefits, such as:

- Not as much storage space is needed to store the file.
- It will take less time to transmit the file from one device to another.
- It will be quicker to upload and download the file.
- Not as much bandwidth is needed to transmit the file over the internet.

There two type of compression that can be used, **lossy** and **lossless**. Each compresses the data in a different way.

Lossy compression uses a compression algorithm that finds the unnecessary and redundant data in the file. This data is permanently removed from the file. This type of compression is mainly used on an image file or a sound file.

Unnecessary data in an image file that could be removed are colours that the human eye cannot distinguish. This means that a lossy compression algorithm for an image file could reduce the size of the file by reducing the colour depth. It could also reduce the size of the file by reducing the image resolution, reducing the number of pixels used to create the image.

Unnecessary data in a sound file that could be removed are sounds that cannot be heard by the human ear. It could also be the removal of softer sounds that are played at the same time as louder sounds, as humans will only hear the louder sound. The method that is used to find these kinds of sounds in a sound file is called **perceptual music shaping**. This is the most common type of method that is used in the file format MP3. The size of a sound file can also be reduced by reducing the sample rate and the sample resolution.

KEY WORDS

compression: a method that uses an algorithm to reduce the size of a file.

lossy: a compression method that reduces the size of a file by permanently removing data.

lossless: a compression method that reduces the size of a file by temporarily altering the data.

KEY WORD

perceptual music shaping: a process that is used in lossy compression that removes sounds that are not normally heard by the human ear.

Questions

7 Why do you think lossy compression is not commonly used on text files?

8 Will lossy compression affect the quality of an image? Why?

9 Will lossy compression affect the quality of a sound recording? Why?